

FamScript Specification Sheet

Mermaid-Based Genealogy & Relationship Diagram Framework

Product Information

Field	Details
Product Name	FamScript
Product Category	Relationship Visualization Framework
Version	1.0
Status	Draft
Prepared By	BuildCraft Labs LLP
Date	May 2026
Framework Type	Declarative Diagram Rendering Engine
Inspired By	Mermaid
Primary Domain	Genealogy & Human Relationship Mapping

1. Executive Summary

FamScript is a Mermaid-inspired declarative framework designed specifically for creating structured genealogy diagrams, relationship graphs, organization charts, and human connection systems.

Unlike generic graph engines, FamScript introduces relationship-aware rendering rules that maintain visual hierarchy, generational balance, semantic clarity, and scalable structure.

The framework aims to become the industry-standard language for programmable human relationship visualization.

2. Product Vision

FamScript aims to provide a universal framework where human relationships can be:

- Programmatically defined
- Visually structured
- Emotionally understandable
- Technically scalable
- Universally readable

The platform should empower:

- Developers
- Families
- Historians
- Researchers
- Enterprises
- Community organizations

to create clean relationship systems using simple text-based syntax.

3. Problem Statement

Current Industry Limitations

Existing graph frameworks such as Mermaid and traditional graph engines face major limitations when modeling genealogy and relationship systems.

Existing Challenges

Problem	Impact
Crooked relationship lines	Poor readability
Uneven node balancing	Visual inconsistency
Chaotic graph expansion	Difficult navigation
Weak spouse alignment	Confusing hierarchy
Lack of generation awareness	Broken genealogy structure
No relationship semantics	Generic disconnected graphs
Organic force-layout rendering	Non-professional appearance

4. Product Objectives

Primary Objectives

Structured Genealogy Rendering

Create clean and professionally balanced relationship layouts.

Declarative Syntax

Allow relationships to be defined using simple human-readable scripts.

Mermaid Compatibility

Retain familiarity for existing Mermaid users while extending relationship capabilities.

Relationship Semantics

Provide first-class support for family and social relationship definitions.

Hierarchical Rendering

Ensure visually stable and structured layouts.

5. Core Design Principles

5.1 Central Character Anchoring

The selected or primary individual must remain visually centered.

Benefits

- Better readability
 - Stable navigation
 - Relationship focus retention
 - Reduced layout drift
 - Predictable expansion
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5.2 Generational Hierarchy Rules

Generation Level	Placement
Ancestors	Above
Current Generation	Center
Descendants	Below

Rendering Requirements

- Automatic generation detection
 - Vertical hierarchy enforcement
 - Balanced expansion logic
 - Symmetrical lineage rendering
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5.3 Structured Rendering Philosophy

FamScript prioritizes:

- Straight relationship lines
- Sharp edge routing
- Minimal edge crossing
- Predictable spacing
- Professional hierarchy flow
- Readable clusters

Visual Inspiration

- Traditional genealogy charts
 - Enterprise org structures
 - Historical lineage systems
 - Professional hierarchy diagrams
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6. Relationship System Architecture

6.1 Family Relationships

Relationship Type	Supported
Parent	Yes
Child	Yes
Sibling	Yes
Grandparent	Yes
Cousin	Yes
Uncle/Aunt	Yes
In-laws	Yes
Step Relations	Yes
Foster Relations	Yes
Adoptive Relations	Yes
Guardian Relations	Yes

6.2 Romantic Relationships

Relationship	Supported
Married	Yes
Divorced	Yes
Engaged	Yes
Separated	Yes
Widowed	Yes
Partnered	Yes
Poly Relationships	Yes

6.3 Social Relationships

Relationship	Supported
Friend	Yes
Mentor	Yes
Caregiver	Yes
Community Member	Yes
Guardian	Yes

6.4 Professional Relationships

Relationship	Supported
Manager	Yes
Employee	Yes
Co-Founder	Yes
Advisor	Yes
Reporting Structure	Yes
Team Hierarchy	Yes

7. Identity & Demographic Framework

Gender Support

FamScript supports inclusive identity representation.

Category	Supported
Male	Yes
Female	Yes
Non-Binary	Yes
Transgender	Yes

Unknown	Yes
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Custom Labels	Yes
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Biological Sex Representation

Category	Supported
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Male	Yes
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Female	Yes
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Intersex	Yes
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Unknown	Yes
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Optional Disclosure	Yes
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Life Status

Status	Supported
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Living	Yes
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Deceased	Yes
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Missing	Yes
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Unknown	Yes
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Relationship Certainty

Type	Meaning
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Confirmed	Verified relationship
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Assumed	Likely but unverified
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Disputed	Conflicting records
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Historical Estimate	Estimated lineage
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8. Deceased & Unknown Individual Support

Deceased Representation

Potential rendering styles:

- Cross markers
 - Dimmed nodes
 - Alternate border patterns
 - Memorial overlays
 - Timeline indicators
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Unknown Individuals

Placeholder nodes supported for:

- Unknown parents
 - Anonymous lineage
 - Historical gaps
 - Missing records
 - Incomplete archives
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9. Rendering Engine Requirements

Layout Engine Requirements

The rendering engine must:

- Keep primary nodes centered
 - Reduce line crossing
 - Prevent overlap
 - Dynamically rebalance layouts
 - Handle large-scale graphs efficiently
 - Support progressive expansion
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Rendering System Requirements

CSS Rendering Goals

Requirement	Priority
Sharp edges	High
Straight lines	High
Consistent spacing	High
Responsive rendering	High
Scalable typography	Medium
Node alignment	High

Layout Philosophy

FamScript explicitly avoids:

- Organic force-directed chaos
 - Random graph movement
 - Unstructured node drift
 - Circular relationship clutter
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10. Mermaid Ecosystem Compatibility

FamScript should maintain compatibility with:

Integration	Support Goal
Markdown embedding	Yes
Static exports	Yes
Web rendering	Yes
Plugin architecture	Planned
Documentation systems	Yes

11. Syntax Framework Direction

Conceptual Example

person A [Atharva]
person B [Father]
person C [Mother]

B -> parent_of -> A
C -> parent_of -> A

A -> married_to -> D
A -> friend_of -> E

Syntax Design Goals

- Human-readable
 - Minimal verbosity
 - Semantic relationships
 - Extensible architecture
 - Easy parsing
 - Markdown-friendly
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12. Technical Specifications

Performance Goals

Metric	Target
Initial Render	< 1 second
Graph Capacity	10,000+ nodes
Zoom Performance	Smooth
Mobile Support	Yes
Dynamic Expansion	Supported

Architecture Goals

Component	Requirement
Parser Engine	Declarative parsing
Rendering Engine	Hierarchical
Layout System	Constraint-based
Export System	Multi-format
API Layer	Extensible

13. Accessibility Requirements

FamScript must support:

- Screen readers
 - Keyboard navigation
 - High contrast themes
 - Semantic relationship descriptions
 - Accessible labels
 - Responsive typography
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14. Export & Output Support

Planned Export Formats

Format	Support
SVG	Yes
PNG	Yes
PDF	Yes
Interactive HTML	Yes
JSON Graph Schema	Yes

15. Developer Experience Goals

FamScript should be:

- Easy to learn
 - Easy to integrate
 - Documentation-friendly
 - Open-standard oriented
 - API extensible
 - Framework agnostic
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16. Primary Use Cases

Family Systems

- Family trees
 - Heritage archives
 - Community lineage
 - Marriage mapping
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Enterprise Systems

- Organization charts
 - Startup structures
 - Team reporting systems
 - Founder relationships
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Historical & Academic

- Royal lineage
 - Historical dynasties
 - Genealogy research
 - Cultural preservation
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17. Non-Goals (Version 1)

The initial release will NOT focus on:

Feature	Status
AI relationship prediction	Excluded
3D rendering	Excluded
Real-time collaboration	Excluded
VR visualization	Excluded
Social networking	Excluded

18. Product Roadmap

Phase 2

Planned Features

- Interactive editing
 - Relationship validation
 - Automatic balancing
 - Smart spacing engine
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Phase 3

Planned Features

- AI-assisted genealogy
 - Timeline integration
 - Historical reconstruction
 - Live collaboration
 - Intelligent lineage detection
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19. Strategic Importance

FamScript is intended to become the foundational relationship rendering engine for:

- Genealogy platforms
- Community systems
- Enterprise hierarchy visualization
- Relationship-centric applications
- Historical preservation systems

The framework can evolve into an independent open ecosystem.

20. Competitive Advantages

Capability	FamScript	Generic Graph Engines
Generation Awareness	Yes	Limited
Relationship Semantics	Native	Manual
Genealogy Layout Rules	Native	No
Central Anchoring	Yes	Rare
Structured Family Rendering	Yes	Weak
Cultural Relationship Support	Yes	Minimal
Human-readable Syntax	Yes	Partial

21. Final Product Philosophy

FamScript is designed around one principle:

Human relationships should be programmable, visual, and universally understandable.

The framework prioritizes:

- Clarity over complexity
- Structure over chaos
- Readability over decoration
- Semantic meaning over raw graph rendering

FamScript aims to make relationship systems emotionally understandable while remaining technically scalable.

22. Recommended Future Enhancements

Suggested Advanced Features

Smart Layout Intelligence

- Automatic branch balancing
- Dynamic generation compression
- Smart spouse alignment

Cultural Relationship Packs

- Indian family systems
- Royal lineage standards
- Clan relationship structures

Advanced Visualization

- Timeline overlays
- Relationship heatmaps
- Interactive lineage playback

Developer Tooling

- VSCode extension
 - Live syntax validator
 - Graph debugger
 - CLI renderer
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23. Suggested Technology Stack

Layer	Recommendation
Rendering Engine	SVG + Canvas Hybrid
Parser	TypeScript
Layout Engine	Constraint-based Graph Engine
Export Engine	SVG/PDF Renderer
Frontend	React + TypeScript
Styling	CSS Variables + Tailwind
Storage Schema	JSON Graph Model

24. Success Metrics (KPIs)

KPI	Target
Render Accuracy	95%+
Large Graph Stability	10k+ nodes
Syntax Learning Time	< 15 minutes
Export Fidelity	High
Mobile Responsiveness	Full Support
Accessibility Compliance	WCAG Compatible

25. Conclusion

FamScript introduces a next-generation structured relationship visualization framework built specifically for genealogy and human connection systems.

By combining:

- Declarative syntax
- Semantic relationship modeling
- Hierarchical rendering
- Professional visual structure
- Mermaid-inspired usability

FamScript has the potential to become the standard framework for relationship-centric visualization systems across genealogy, enterprise, academic, and social domains.